APM5557: Partial Differential Equations
The Gelfand Triple

A Gelfand triple (also called a *rigged Hilbert space*) is a chain

$$V \subset H \subset V'$$

of a "nice" space V, a Hilbert space H, and the dual V' of V. **The middle space H acts as a pivot: its inner product is extended to a pairing between V' and V, so that objects too rough to lie in H (such as δ or $-\Delta u$ for $u \in H_0^1$) can still be "integrated against" elements of V.** This is the setting for the weak formulation of elliptic and parabolic PDEs.

1 Definition

Definition. Let H be a Hilbert space with inner product $(\cdot,\cdot)_H$, and let V be a reflexive Banach space (in practice, usually a Hilbert space) such that

(i) $V \subset H$ with continuous embedding: $\|v\|_H \le C\|v\|_V$ for all $v \in V$;

(ii) V is dense in H.

Identify H with its dual H' using the Riesz representation theorem. Then

$$V \hookrightarrow H \cong H' \hookrightarrow V',$$

and the triple (V, H, V') is called a *Gelfand triple*. H is called the *pivot space*.

How H sits inside V'

Each $h \in H$ defines a functional on V by

$$\ell_h(v) = (h, v)_H, \qquad v \in V.$$

This functional is bounded on V, because $|(h,v)_H| \le \|h\|_H \|v\|_H \le C\|h\|_H \|v\|_V$. So $\ell_h \in V'$ with $\|\ell_h\|_{V'} \le C\|h\|_H$.

The map $h \mapsto \ell_h$ is **injective**: if $(h,v)_H = 0$ for all $v \in V$, then by density of V in H it holds for all $v \in H$, so $h = 0$. This is exactly where the density assumption (ii) is used. We identify h with ℓ_h and write $H \subset V'$. If V is reflexive, H is also dense in V'.

The duality pairing extends the inner product

Write $\langle f, v\rangle = \langle f, v\rangle_{V',V}$ for the action of $f \in V'$ on $v \in V$. Under the identification above,

$$\boxed{\langle h, v\rangle = (h,v)_H \quad \text{whenever } h \in H,\ v \in V.}$$

So $\langle\cdot,\cdot\rangle$ is "the H inner product, extended to the case where the first argument is rougher than H and the second is smoother."

2 The canonical example: $H_0^1(\Omega) \subset L^2(\Omega) \subset H^{-1}(\Omega)$

Take $\Omega \subset \mathbb{R}^n$ bounded and

$$V = H_0^1(\Omega), \qquad H = L^2(\Omega), \qquad V' = H^{-1}(\Omega).$$

- The embedding is continuous: $\|v\|_{L^2} \le \|v\|_{H^1}$. Density holds because $C_c^\infty(\Omega) \subset H_0^1$ is already dense in L^2.
- By the Poincaré inequality we may use $\|v\|_{H_0^1} = \|\nabla v\|_{L^2}$ as the norm on V.
- The norm on H^{-1} is the dual norm

$$\|f\|_{H^{-1}} = \sup_{0 \neq v \in H_0^1} \frac{|\langle f, v\rangle|}{\|\nabla v\|_{L^2}}.$$

- Every $f \in H^{-1}$ can be written $f = g_0 + \sum_{i=1}^n \partial_i g_i$ with $g_0, \dots, g_n \in L^2$, acting by $\langle f, v\rangle = \int g_0 v - \sum_i \int g_i \, \partial_i v$.

Elements of H^{-1} that are not in L^2

1. **Derivatives of L^2 functions.** If $g \in L^2(\Omega)$, then $\partial_i g \in H^{-1}$, with $\langle \partial_i g, v\rangle = -\int g \, \partial_i v$. For example, the derivative of the Heaviside function, $H' = \delta$, lies in $H^{-1}(-1, 1)$.
2. **Point masses in 1D.** On an interval (a, b), functions in H_0^1 are continuous with $|v(x_0)| \le \sqrt{b-a}\,\|v'\|_{L^2}$, so $\langle \delta_{x_0}, v\rangle = v(x_0)$ is bounded and $\delta_{x_0} \in H^{-1}(a, b)$.
3. **But not in 2D or higher.** For $n \ge 2$, H_0^1 contains unbounded functions (e.g. $\log\log(e/|x|)$ in 2D), so point evaluation is not bounded and $\delta \notin H^{-1}$.
4. **Laplacians of H_0^1 functions.** For $u \in H_0^1$, $-\Delta u \in H^{-1}$ with $\langle -\Delta u, v\rangle = \int \nabla u \cdot \nabla v$.

3 A concrete picture using Fourier sine series

On $\Omega = (0, \pi)$ the Dirichlet eigenfunctions of $-d^2/dx^2$ are

$$\varphi_n(x) = \sqrt{2/\pi}\,\sin(nx), \qquad \lambda_n = n^2,$$

an orthonormal basis of $L^2(0, \pi)$. Writing $v = \sum c_n \varphi_n$, the three spaces become weighted sequence spaces:

Space	Condition on coefficients	Squared norm
$H_0^1(0,\pi)$	$\sum n^2 \lvert c_n\rvert^2 < \infty$	$\|v'\|_{L^2}^2 = \sum n^2 \lvert c_n\rvert^2$
$L^2(0,\pi)$	$\sum \lvert c_n\rvert^2 < \infty$	$\|v\|_{L^2}^2 = \sum \lvert c_n\rvert^2$
$H^{-1}(0,\pi)$	$\sum n^{-2} \lvert c_n\rvert^2 < \infty$	$\|f\|_{H^{-1}}^2 = \sum n^{-2} \lvert c_n\rvert^2$

The pairing of $f = \sum f_n \varphi_n \in H^{-1}$ with $v = \sum c_n \varphi_n \in H_0^1$ is

$$\langle f, v\rangle = \sum_n f_n c_n = \sum_n \big(n^{-1} f_n\big)\big(n\, c_n\big),$$

which converges by Cauchy–Schwarz: the weight n^{-1} lost by f is made up by the weight n gained by v.

Example (δ_{x_0})**.** Its coefficients are $f_n = \langle \delta_{x_0}, \varphi_n \rangle = \sqrt{2/\pi}\sin(nx_0)$. These are bounded but do not decay, so $\sum |f_n|^2 = \infty$ (not in L^2) while $\sum n^{-2}|f_n|^2 < \infty$ (in H^{-1}).

The same picture gives the whole scale H^s, $s \in \mathbb{R}$, with norm $\sum n^{2s}|c_n|^2$, and a Gelfand triple $H^s \subset L^2 \subset H^{-s}$ for every $s > 0$.

4 Why we don't also identify V with V'

If V is itself a Hilbert space, Riesz gives an isomorphism $R : V \to V'$, $\langle Ru, v \rangle = (u, v)_V$. It is tempting to identify V with V' too, but **you cannot use both identifications at once**.

For $V = H_0^1$ with $(u, v)_V = \int \nabla u \cdot \nabla v$, the Riesz map is

$$R = -\Delta : H_0^1 \to H^{-1}.$$

Identifying V with V' would identify u with $-\Delta u$, whereas the pivot identification $H \cong H'$ identifies u with u itself. These contradict each other. On $(0, \pi)$ (see the previous section), $-\Delta\varphi_2 = 4\varphi_2$, so the first identification would send φ_2 to $4\varphi_2 \in V'$, while the second sends it to $\varphi_2 \in V'$. Using both would force $\varphi_2 = 4\varphi_2$. A Gelfand triple makes a choice: **only the pivot space H is identified with its dual**, and V' is kept as a genuinely larger space.

5 Application: weak formulation of elliptic problems

Consider $-\Delta u = f$ in Ω, $u = 0$ on $\partial\Omega$, with $f \in H^{-1}$.

Theorem. *For every $f \in H^{-1}(\Omega)$ there is a unique $u \in H_0^1(\Omega)$ with*

$$\int_\Omega \nabla u \cdot \nabla v = \langle f, v \rangle \qquad \text{for all } v \in H_0^1(\Omega),$$

and $\|\nabla u\|_{L^2} = \|f\|_{H^{-1}}$.

Sketch. The left side is the inner product of H_0^1, and $v \mapsto \langle f, v \rangle$ is a bounded linear functional on H_0^1. The Riesz representation theorem gives a unique u, with norm equal to the functional's norm. (For a general coercive bilinear form, use Lax–Milgram instead.) □

So $-\Delta : H_0^1 \to H^{-1}$ is an isometric isomorphism. The Gelfand triple is what lets us state this cleanly:

- the solution lives in V;
- the data lives in V', which allows $f = \delta$ in 1D, or $f = \partial_i g$ with $g \in L^2$;
- when the data happens to lie in $H = L^2$, the pairing reduces to $\langle f, v \rangle = \int fv$, which is the usual weak form.

6 Application: evolution equations

For the heat equation $u_t - \Delta u = f$, the natural solution space is

$$W(0, T) = \{u \in L^2(0, T; V) : u' \in L^2(0, T; V')\},$$

with $V = H_0^1$, $V' = H^{-1}$. The time derivative is allowed to be rough (in V') because $u_t = \Delta u + f$ and $\Delta u \in H^{-1}$.

Theorem (Lions–Magenes)**.** *If $u \in W(0,T)$, then $u \in C([0,T];H)$ (after modification on a null set), and*

$$\frac{d}{dt}\|u(t)\|_H^2 = 2\langle u'(t), u(t)\rangle \quad \textit{for a.e. } t.$$

This is the rigorous form of the energy identity. It shows that the initial condition $u(0) = u_0 \in H$ makes sense, and it is used to prove uniqueness and energy estimates such as $\|u(t)\|_{L^2}^2 + 2\int_0^t \|\nabla u\|_{L^2}^2\, ds \le \|u_0\|_{L^2}^2$ when $f = 0$.

7 Other Gelfand triples

Triple	Pivot	Where it is used
$H_0^1 \subset L^2 \subset H^{-1}$	L^2	Dirichlet problems, heat equation
$H^1 \subset L^2 \subset (H^1)'$	L^2	Neumann problems ($V = H^1$ is dense in L^2, but $(H^1)' \neq H^{-1}$)
$H^s \subset L^2 \subset H^{-s}$	L^2	higher-order and fractional operators
$\mathcal{S} \subset L^2(\mathbb{R}^n) \subset \mathcal{S}'$	L^2	Fourier analysis; quantum mechanics, where plane waves e^{ikx} and δ_{x_0} are "generalized eigenfunctions" of momentum and position

The last one is not a Gelfand triple in the Hilbert-space sense ($\mathcal{S}$ is a Fréchet space, not a Hilbert space), but the structure is the same and it is the original setting of Gelfand's "rigged Hilbert space."

8 Summary

- A Gelfand triple $V \subset H \subset V'$ needs V continuously and densely embedded in a Hilbert space H.
- Only H is identified with its dual. V' is strictly larger than H and contains rougher objects.
- The pairing $\langle f, v\rangle_{V',V}$ extends $(f,v)_H$.
- Differential operators like $-\Delta$ map V isomorphically onto V', which is the functional-analytic content of the weak formulation.

A History

The Gelfand triple is named after Israel Gelfand. It is sometimes attributed to John von Neumann, but his work came earlier and supplied the Hilbert-space framework that Gelfand's construction extends. **Von Neumann's own approach was designed to avoid exactly the objects that Gelfand triples make room for.**

John von Neumann (around 1927–1932)

- He put quantum mechanics on a rigorous Hilbert-space footing in *Mathematical Foundations of Quantum Mechanics* (1932).
- He developed the spectral theory of unbounded self-adjoint operators.
- He openly criticized Dirac's use of δ functions and of "eigenvectors" that aren't in the Hilbert space, such as plane waves e^{ikx}. He replaced them with projection-valued measures, so in his framework every object stays inside H.

Israel Gelfand and collaborators (mid-1950s to early 1960s)

- They introduced the rigged Hilbert space $\Phi \subset H \subset \Phi'$, where Φ is a nuclear space. A key step was the nuclear spectral theorem, from Gelfand and Kostyuchenko (1955). Krzysztof Maurin proved a similar result independently, so it is sometimes called the Gelfand–Maurin theorem.
- It was developed fully in *Generalized Functions*, Vol. 4, by Gelfand and Vilenkin (1961).
- The theorem says that a self-adjoint operator has a complete set of generalized eigenvectors in Φ'. That made Dirac's bra-ket formalism rigorous, which is the opposite of von Neumann's strategy of avoiding it.
- The word "rigged" is a translation of the Russian *osnashchyonnoe*, meaning "equipped."

Other contributors

- **Laurent Schwartz** created distribution theory (1945–1950), which gave $\mathcal{S} \subset L^2 \subset \mathcal{S}'$ its outer spaces.
- **Jacques-Louis Lions and Enrico Magenes** made the Hilbert-space version $V \subset H \subset V'$ the standard tool for weak formulations of PDEs, which is the version used in this document. The term "pivot space" (*espace pivot*) comes from that French school.
- **Arno Böhm, J. E. Roberts, and J.-P. Antoine** brought rigged Hilbert spaces into quantum physics in the 1960s.

Summary

The line of influence runs from von Neumann's Hilbert space, through Schwartz's distributions, to Gelfand's rigged Hilbert space and then Lions's Gelfand triples for PDEs. Section 7 calls $\mathcal{S} \subset L^2 \subset \mathcal{S}'$ the original setting of Gelfand's rigged Hilbert space. Strictly, Gelfand worked with general nuclear spaces Φ, and $\mathcal{S}$ is the standard example of one rather than the only setting.